21-2 LIGHT-EMITTING DIODES (LED)

LED Operation and Construction

Charge carrier recombination occurs at a forward-biased *pn*-junction as electrons cross from the *n*-side and recombine with holes on the *p*-side. Free electrons have a higher energy level than holes, and some of this energy is dissipated in the form of heat and light when recombination takes place. If the semiconductor material is translucent, the light is emitted and the junction becomes a light source, that is, a *light-emitting diode (LED)*.

A cross-sectional view of an LED junction is shown in Fig. 21-2a. The semiconductor material is gallium arsenide (GaAs), gallium arsenide phosphide (GaAsP), or gallium phosphide (GaP). An *n*-type epitaxial layer is grown upon a substrate, and the *p*-region is created by diffusion. Since charge carrier recombinations occur in the *p*-region, the *p*-region is kept uppermost to allow the light to escape. The metal film anode connection is

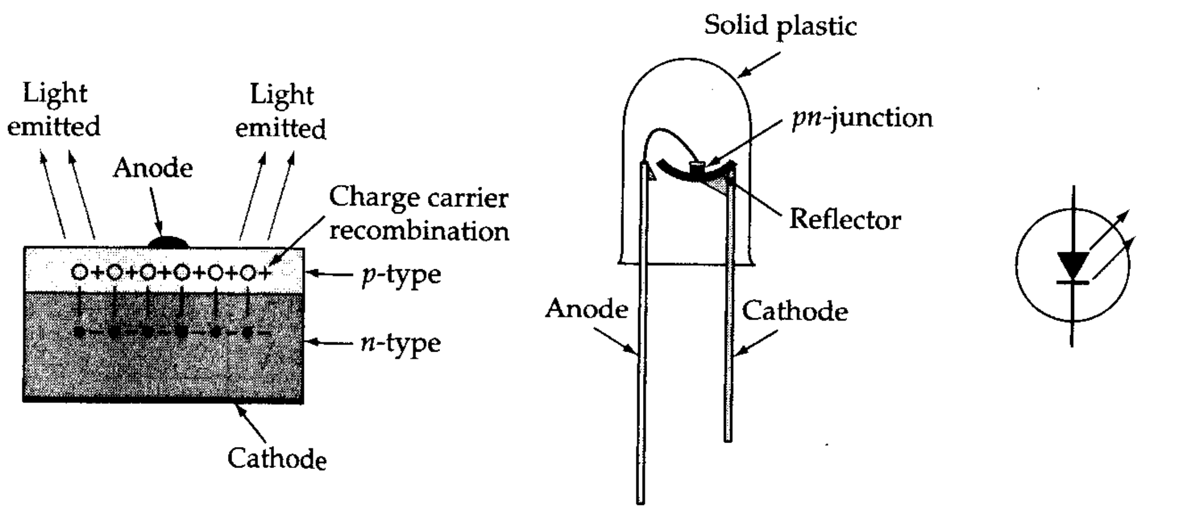


Figure 21-2 A light-emitting diode (LED) produces energy in the form of light when charge carrier recombination occurs at the *pn*-junction.

patterned to allow most of the light to be emitted. A gold film is applied to the bottom of the substrate to reflect as much light as possible toward the surface of the device and to provide a cathode connection. LEDs made from GaAs emit infrared (invisible) radiation. GaAsP material produces either red light or yellow light, while red or green light can be created by using GaP. Through the use of various materials, LEDs can be manufactured to produce light of virtually any color.

Figure 21-2b shows the typical construction of a LED. The *pn*-junction is mounted on a cup-shaped reflector, wires are provided for anode and cathode connection, and the device is encapsulated in an epoxy lens. The lens can be colorless or colored, and (when not energized), the lens color identifies the LED light color. The color of the light emitted by the energized LED is determined solely by the *pn*-junction material. Some LEDs have glass particles embedded in the epoxy lens to diffuse the emitted light and increase the viewing angle of the device. The LED circuit symbol is shown in Fig. 21-2c. Note that the arrow directions indicate emitted light.

Characteristics and Parameters

LED characteristics are similar to those of other semiconductor diodes, except that (as shown in the partial specification in Fig. 21-3) the typical forward voltage drop is 1.6 V. Note also that the reverse breakdown voltage can be as low as 3 V. In some circuits it is necessary to include a diode with a high reverse breakdown voltage in series with a LED. The forward current used with a LED is usually in the 10 mA to 20 mA range, but (depending on the particular device) the peak current can be as high as 90 mA. LED luminous intensity depends on the forward current level; it is usually specified at 20 mA. The peak wavelength of the light output is also normally listed on the specification.

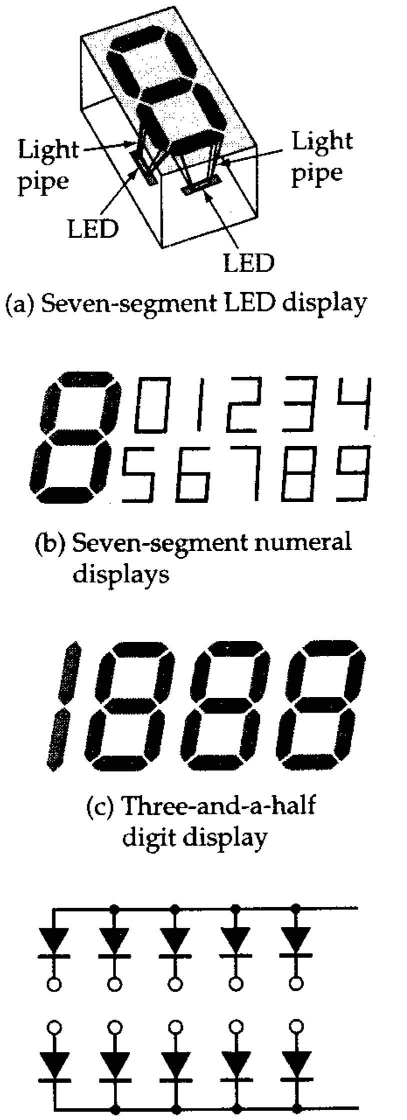
21-3 SEVEN-SEGMENT DISPLAYS

LED Seven-Segment Display

The arrangement of a *seven-segment* LED numerical display is shown in Fig. 21-6a. Since the actual LED devices are very small, solid plastic *light pipes* are often employed to enlarge the lighted surface, as shown. Any numeral from 0 to 9 can be indicated by passing current through the appropriate segments (Fig. 21-6b). Fig. 21-6(c) shows three seven-segment displays together with a two-segment display referred to as a *half-digit*. The whole display, termed a *three-and-a-half digit display*, can be used to indicate numerical values up to a maximum of 1999.

The LEDs in a seven-segment display may be connected in *common-anode* or in *common-cathode* configuration (Fig. 21-6d). When an LED seven-segment display is selected, it is important to determine which of the two connecting arrangements is required.

The relatively large amounts of current consumed by LED seven-segment displays are their major disadvantage. Apart from this, LEDs have the advantage of long life and ruggedness.



(a) Seven-segment LED display

(b) Seven-segment numeral displays

(c) Three-and-a-half digit display

(d) Common-anode and common-cathode connections

Figure 21-6 Light-emitting diodes can be arranged in seven-segment format for numerical displays.

Example 21-4

Calculate the total power supplied to a $3^1/_2$ digit LED display when it indicates 1888. A 5 V supply is used, and each LED has a 10 mA current.

Solution

Total segments: $N = [3 \times (\text{segments for } 8)] + [1 \times (\text{segments for } 1)]$

$$= (3 \times 7) + (1 \times 2) = 23$$

Total current: $I_T = N \times (\text{current per segment}) = 23 \times 10\ \text{mA}$

$= 230\ \text{mA}$

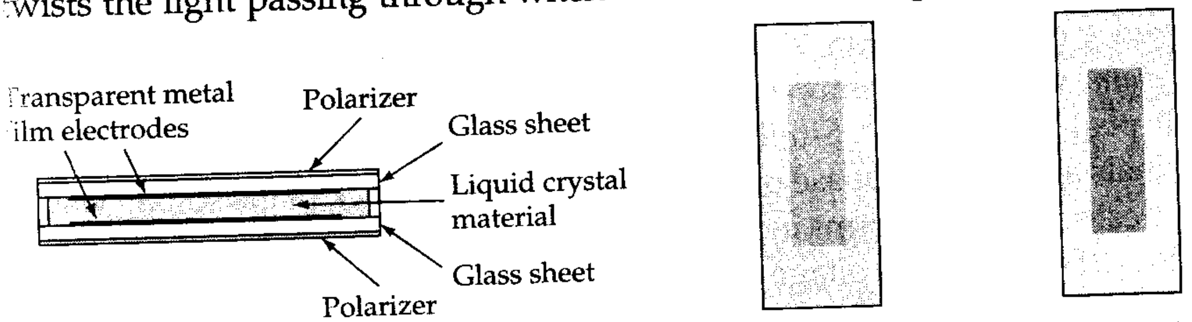
Power: $P = I_T \times V_{CC} = 230\ \text{mA} \times 5\ \text{V}$

$= 1.15\ \text{W}$

Liquid Crystal Cells

Liquid crystal material is a liquid that exhibits some of the properties of a solid. The molecules in ordinary liquids normally have random orientations. In liquid crystals, however, the molecules are oriented in a definite crystal pattern.

A liquid crystal cell consists of a very thin layer of liquid crystal material sandwiched between glass sheets, as illustrated in Fig. 21-7a. The glass sheets have transparent metal film electrodes deposited on the inside surfaces. In the commonly used *twisted nematic* cell, two thin, polarizing optical filters are placed at the surface of each glass sheet. The liquid crystal material employed twists the light passing through when the cell is not energized. This twisting

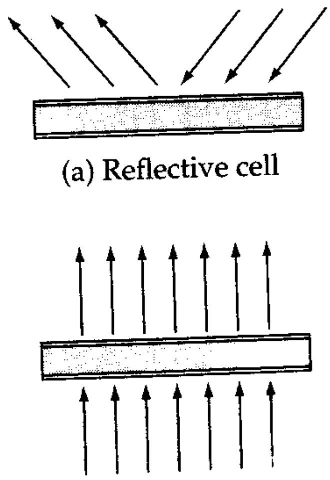


(a) Cross-section of a liquid crystal cell (b) Unenergized cell (c) Energized cell

Figure 21-7 **A liquid crystal cell consists of a layer of liquid crystal material sandwiched between glass sheets with transparent metal film electrodes and polarizers.**

allows the light to pass through the polarizing filters, so that the cell is semi-transparent. When energized, the liquid molecules are reoriented so that no twisting occurs, and no light can pass through. Thus, the energized cell can appear dark against a bright background (see Figs 21-7b and c). The cells can also be manufactured to appear bright against a dark background.

Seven-segment numerical (and other type) displays made from liquid crystal cells are referred to as *liquid crystal displays (LCDs)*. Two types of LCDs are illustrated in Fig. 21-8. The *reflective*-type shown in Fig. 21-8a relies on reflected light. The cell is placed on a reflective surface, so that when not energized it is just as reflective as the surrounding material; consequently, it disappears. When the cell is



(a) Reflective cell

(b) Transmittive cell

Figure 21-8 **Reflective-type liquid crystal cells reflect incident light. Transmittive-type cells pass light from behind.**

energized, no light is reflected from it, and it appears dark against the bright background. The *transmittive* cell in Fig. 21-8b allows light to pass through from the back of the cell when it is not energized. When the cell is energized, the light is blocked, and here again the cell appears dark against a bright background. The *trans-reflective* cell is a combination of transmittive and reflective types.

LCD Seven-Segment Displays

Because liquid crystal cells are light reflectors or transmitters rather than light generators, they consume very small quantities of energy. The only energy required by the cell is that needed to activate the liquid crystal. The total current flow through four small seven-segment LCDs is usually about 20 μA. However, LCDs require an ac voltage supply, in the form of either a sine wave or a square wave. This is because a continuous direct current produces a plating of the cell electrodes that could damage the device. Repeated reversal of the current prevents this problem.

A typical LCD supply is a 3 V to 8 V peak-to-peak square wave with a frequency of 60 Hz. Figure 21-9 illustrates the square-wave drive method. The *back plane*, which is common to all of the cells, is supplied with a square wave (with peak voltage V_P). Similar square waves applied to each of the other terminals are either in phase or in antiphase with the back plane square wave. Those cells with waveforms in phase with the back plane waveform (cells e and f in Fig. 21-9) have no voltage developed across them. Since both terminals of the segment are at the same potential, they are not energized. The cells with square waves in antiphase with the back plane input have a square wave with peak voltage $2V_P$ developed across them, and consequently, they are energized.

Unlike LED displays, which are usually quite small, LCDs can be fabricated in almost any convenient size. The major advantage of LCDs is

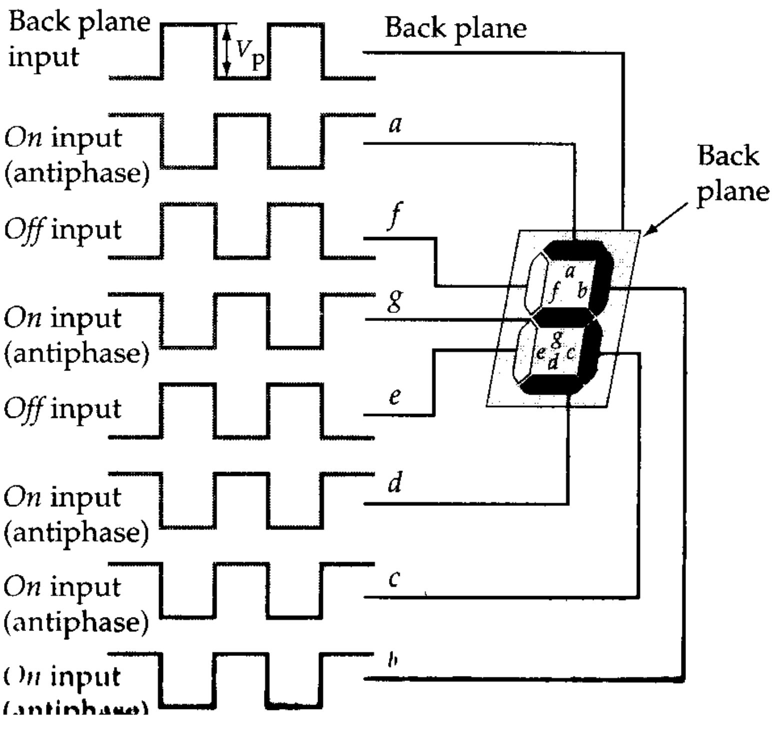


Figure 21-9 Liquid crystal display using a square wave supply. A segment is energized when its input is in antiphase with the back plane input.

their low power consumption. Perhaps the major disadvantage of the LCD is its decay time of 150 ms (or more). This is very slow compared to the rise and fall times of LEDs. In fact, the human eye can sometimes observe the fading out of LCD segments switching *off*.